

History of Programming Languages

Programming Languages Intro

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Human Language

- Used by people to communicate ideas and discuss



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Machine Language



Computers are not as complex as humans



Computer features and commands accessed via “code”

Programming Languages



Domain expression is reduced compared to natural languages



Enable communication between

Machine - developer
Developer - developer



Set of syntactic and semantic rules, artificially created

Standardize the instructions

A hint of history

- It all started with Machine Languages and Assembly languages (40's)
 - Close to the way machine operates
 - Use low level commands to manipulate memory and processor

```
68 0x52ac76: movl 7306562(%ebx), %eax
69 0x52ac7c: movl %eax, -20(%ebp)
70 0x52ac7f: movl $0, (%edi,%eax)
71 0x52ac86: testl %esi, %esi
72 0x52ac88: je 0x52ad21 ; -
    [UINavigationController _updateScrollViewFromViewController:
    toViewController:] + 425
73 0x52ac8e: movl 7306542(%ebx), %eax
74 0x52ac94: movl (%edi,%eax), %eax
75 0x52ac97: movl %eax, -24(%ebp)
76 0x52ac9a: movl 7212558(%ebx), %eax
77 0x52aca0: movl %eax, 4(%esp)
78 0x52aca4: movl %esi, (%esp)
79 0x52aca7: calll 0x9bff06 ; symbol stub for:
    objc_msgSend
80 0x52acac: movl %eax, -28(%ebp)
81 0x52acaf: movl %edx, -32(%ebp) Thread 1: instruction step over
82 0x52acb2: movl 7211062(%ebx), %eax
83 0x52acb8: movl %eax, 4(%esp)
84 0x52acbc: movl %esi, (%esp)
```

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A hint of history: the 50's

- High-level languages introduced in the 50's
 - Platform-independent languages (not the binary, though)
 - Easier to communicate with the developers

- FORTRAN, LISP, COBOL

```
***** ***** Top of Data Set *****
000010 //MATE06A JOB (123), 'CBLRUN', CLASS=A, MSG=
000020 //                NOTIFY=&SYSUID
000200 //*****
000300 //* JCL TO RUN A SIMPLE COBOL PROGRAM
000400 //*****
000500 //STEP01 EXEC PGM=GOTO
000600 //STEPLIB DD DSN=MATE06.VKKUMAR.TEST.LIB
000700 //SYSOUT DD SYSOUT=*
000800 //SYSIN DD *
000801 MALE
000802 /*
000810 //
***** ***** Bottom of Data Set *****

F1=Help
F8=Down
```

```
1 ]=> (define (removeadj L) (if (null? L) L
(if (equal? (car L) (cadr L)) (removeadj (cdr L))
(append (list (car L)) (removeadj (cdr L)))))
```

```
;Value: removeadj
```

```
1 ]=> (removeadj '(a b a a c c))
```

```
;The object (), passed as an argument to safe-car, is
;To continue, call RESTART with an option number:
; (RESTART 1) => Return to read-eval-print level 1.
```

```
2 error> RESTART 1
```

A hint of history: Scientific application

FORTRAN (FORMula TRANslator)

- IBM team led by John Backus
- Applications in Engineering
- Still used currently
- Efficiency

LISP (LISt Processing)

- designed for easy manipulation of data strings
- influenced by the notation of Alonzo Church's lambda calculus
- favored programming language for artificial intelligence (AI) research

A hint of history

- COBOL (still late 50's)
 - Better precision and automation of business processes
 - Platform-independent language
 - Inspired by FLOW-MATIC (by Grace Hopper)
 - Payroll, accounting, inventory control

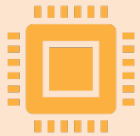


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A hint of history: 60's and 70's



Expansion and **education**



Logo, Pascal, Basic



Pascal

Niklaus Wirth
Professor @ ETH Zurich

A hint of history: 70's - 80's

System programming

- Operating Systems, Network, Virtual machine
- Most implemented in C
 - CPL → BCPL → B → C
 - Mid-level language

Consolidation, modules, performance

- Object-orientation (ADA, C++)
- RISC movement: hardware should be optimized for compilers
- Fifth generation Programming Languages: AI/Logic Programming

A hint of history: 90's

The Internet age

- Integration with web-browser: JavaScript birth
- “Internet” languages: PHP, CGI



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Rapid Application Development

- IDEs
- Java, Object Pascal, Visual Basic

Script languages

- Python, Lua, R

A hint of history: Modern and contemporary

AI and Data
Analysis

“Pure”
Functional
programming

Systems-of-
systems and
components

Concurrent
and distributed
programming

Meta-
programming

Low/No-code

**See you in our next
Lesson!**